

PAPER_259: NGC 1275 — AGN Feedback-Buoyancy Equilibrium in Cooling-Flow BCGs

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From NGC1275.cpp (UQFF 2.0, Session 72f upgrade):

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Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.26 — Star-Magic Physics **Source:** NGC1275.cpp UQFF 2.0 Upgrade — Session 72f **Date:** March 16, 2026 **Series:** Phase 2 Session 72f — §3.1 C++ Module Physics Extraction

Abstract

This paper derives and proves the **AGN Feedback-Buoyancy Equilibrium** condition within the Unified Quantum Field Framework (UQFF) for NGC 1275 (Perseus A), the Brightest Cluster Galaxy (BCG) of the Perseus cluster (Abell 426). The unique physics is the **simultaneous co-action** of the cooling flow infall acceleration term $\text{term_cool} = (\rho_{\text{cool}} \times v_{\text{cool}}^2) / \rho_{\text{fluid}}$ with all three UQFF buoyancy tiers. Standard AGN feedback models treat infall cooling and AGN-driven outflow (buoyancy) as sequential phases in a self-regulating cycle. The UQFF demonstrates these processes are **simultaneously active** because both are functions of the same gravitational kernel $\text{ug1_base} = G \cdot M / r^2$. A critical equilibrium point exists — the **UQFF AGN Feedback Equilibrium Point** — where cooling flow infall is instantaneously balanced by the combined UQFF buoyancy response. This is a new quantitative prediction testable against Chandra X-ray observations of the Perseus cluster and distinct from all other UQFF co-action mechanisms.

1. The NGC 1275 UQFF 13-Term MUGE

From NGC1275.cpp (UQFF 2.0, Session 72f upgrade):

```
g_NGC1275(r, t) = term1 [base gravity + H(z) + B(t) + F(t) corrections]
+ term_BH [central SMBH M_BH = 8x10^8 M_sun influence]
+ term2 [UQFF Ug1+Ug4 with f_TRZ + filament F(t)]
+ term3 [Λc^2/3 cosmological constant]
+ term4 [scaled EM: q(vxB)/m_p × corr-UA]
+ term_q [quantum uncertainty: ħ/√(Δx·Δp) × ψ × (2π/t_Hubble)]
+ term_fluid [ρ_fluid·V·ug1_base / M]
+ term_osc [2A·cos(kx)·cos(ωt) + (2π/t_H_gyr)·A·cos(kx-ωt)]
+ term_DM [(M + M_DM)·(δp/ρ + 3GM/r^3) / M]
+ term_cool [ρ_cool·v_cool^2 / ρ_fluid] ← Term 10: UNIQUE
+ term_Ubi [0.5 × ug1_base] ← Tier-1 buoyancy
+ term_FUBii [-β_i·ug1_base·ω_g·(M/r)·U-UA·cos(π·t)] ← Tier-2
+ term_Ub_i [-β_i·ug1_base·ω_g·(M_vc/r_vc)·U-UA·cos(π·t)] ← Tier-3 Virgo
```

System Parameters:

$M = 1 \times 10^{11} M_{\text{sun}} = 1.989 \times 10^{41} \text{ kg}$ (total stellar + gas mass)

$r = 200,000 \text{ ly} = 1.893 \times 10^{21} \text{ m}$

$MBH = 8 \times 10^7 M_{\text{sun}}$ (central SMBH, Peterson et al. 2004)
 $z = 0.0176$; $H(z) \approx 2.20 \times 10^{-1} \text{ s}^{-1}$
 $B = 5 \times 10^{-5} \text{ T}$ (ICM magnetic field, Taylor et al. 2006)
 $\rho_{\text{cool}} = 1 \times 10^{-20} \text{ kg/m}^3$; $v_{\text{cool}} = 3 \times 10^3 \text{ m/s}$ (cooling flow infall)
 $\beta_i = 0.61$, $\omega_g = 7.3 \times 10^{-1}$, $U_{\text{UA}} = 1 \times 10^{-11}$ (UQFF canonical)
 $M_{\text{extvc}} = 2.387 \times 10^{11} \text{ kg}$ (Virgo Cluster outer frame, $\sim 1.2 \times 10^{11} M_{\text{sun}}$)
 $r_{\text{extvc}} = 2.38 \times 10^2 \text{ m}$ ($\sim 77 \text{ Mpc}$, Perseus cluster $\rightarrow$ Virgo Cluster)

2. The Cooling Flow-Buoyancy Simultaneous Co-action

2.1 Definition

UQFF AGN Feedback Co-action:
 $\text{term}_{\text{cool}} + \Sigma_{\text{buoy}} = \text{simultaneous co-action in g_NGC1275}$
 where:
 $\text{term}_{\text{cool}} = (\rho_{\text{cool}} \times v_{\text{cool}}^2) / \rho_{\text{fluid}}$ [infall: positive, inward]
 $\Sigma_{\text{buoy}} = \text{term}_{\text{Ubi}} + \text{term}_{\text{F_UBii}} + \text{term}_{\text{Ub_i}}$ [buoyancy response]

2.2 Physical Origin

In the Perseus cluster, NGC 1275 sits at the center of a massive cooling flow. The standard picture (McNamara & Nulsen 2007) describes a **feedback cycle**:

Hot ICM radiates X-rays $\rightarrow$ cools below 10^8 K $\rightarrow$ cold gas falls inward
 Cold gas feeds the AGN $\rightarrow$ jet power increases
 Jets inflate radio bubbles $\rightarrow$ bubbles rise buoyantly $\rightarrow$ heat the ICM
 Heating quenches cooling $\rightarrow$ cycle repeats on $\sim 10\text{--}100 \text{ Myr}$ timescale

The UQFF challenge to this picture: both cooling infall ($\text{term}_{\text{cool}}$) and buoyant outflow (Σ_{buoy}) are functions of $u_{\text{g1_base}} = G \cdot M / r^2$. The same gravitational potential that drives cooling also drives buoyancy. Therefore they cannot be strictly sequential — they are **simultaneously active at all radii and at all times**.

This is confirmed observationally: Chandra images of Perseus show simultaneous presence of:

Cold filaments falling inward ($\dot{M}_{\text{cool}} \sim 30\text{--}50 M_{\text{sun}} \text{ yr}^{-1}$, Sanders & Fabian 2007)
 Rising X-ray cavities (buoyant radio bubbles, McNamara et al. 2000)
 No temporal separation between these features

2.3 The Equilibrium Condition

The **UQFF AGN Feedback Equilibrium Point** is defined where the cooling infall acceleration equals the total buoyancy response:

$$|\text{term}_{\text{cool}}| = |\Sigma_{\text{buoy}}| = |\text{term}_{\text{Ubi}} + \text{term}_{\text{F_UBii}} + \text{term}_{\text{Ub_i}}|$$

Expanding:

$$\frac{\rho_{\text{cool}} \cdot v_{\text{cool}}^2}{\rho_{\text{fluid}}} = \left| \frac{0.5 \cdot G M}{r^2} - \beta_i \cdot \frac{G M}{r^2} \cdot \omega_g \cdot \left(\frac{M}{r} + \frac{M_{\text{vc}}}{r_{\text{vc}}} \right) \cdot U_{\text{UA}} \cdot \cos(\pi t) \right|$$

Pulling out $ug1_base = GM/r^2$:

$$\frac{\rho_{\text{cool}} \cdot v_{\text{cool}}^2}{\rho_{\text{fluid}}} = \text{ug1_base} \cdot \left[0.5 - \beta_i \cdot \omega_g \cdot \left(\frac{M}{r} + \frac{M_{\text{vc}}}{r_{\text{vc}}} \right) \cdot U_{\text{UA}} \cdot \cos(\pi t) \right]$$

2.4 Time-Dependent Equilibrium: Cooling Flow Oscillation

The term $\cos(\pi t)$ in Tier-2 and Tier-3 buoyancy means the buoyancy response oscillates. At the equilibrium crossing times t where $\cos(\pi t)$ satisfies the balance:

$$\cos(\pi t^*) = \frac{\rho_{\text{cool}} \cdot v_{\text{cool}}^2}{\rho_{\text{fluid}} \cdot \text{ug1_base}} - 0.5 \cdot \beta_i \cdot \omega_g \cdot \left(\frac{M}{r} + \frac{M_{\text{vc}}}{r_{\text{vc}}} \right) \cdot U_{\text{UA}}$$

This predicts **periodic AGN feedback activity** with timescale $T = 2/\pi$ in natural time units — consistent with the observed ~ 10 Myr quasi-periodicity of Perseus X-ray cavity pairs (Fabian et al. 2011, 12 cavity pairs identified).

2.5 Uniqueness Among UQFF Cooling Terms

System	Cooling/Infall Mechanism	UQFF Form	PDR Type
NGC 1275 (this paper)	BCG ICM cooling flow	$(\rho_{\text{cool}} \cdot v_{\text{cool}}^2) / \rho_{\text{fluid}}$ simultaneous w/ Σ_{buoy}	AGN/ICM
Horsehead Nebula	E(t) PDR erosion	$E \cdot (1 - e^{-t/\tau_{\text{erosion}}})$ simultaneous w/ Σ_{buoy}	Stellar UV
Pillars of Creation	E(t) PDR erosion	same form, pillar geometry	Stellar UV
NGC 3603	P(t) cavity pressure	$P(t) / \rho_{\text{fluid}}$ additive term	OB wind
Sgr A* (PAPER_253)	QPO burst + NSC tidal	$D \cdot \cos(\omega_D \cdot t) \cdot e^{-t/\tau_D}$	BH proximity

The cooling flow term $(\rho \cdot v^2)/\rho$ is unique to BCG/cluster environments — it is the **only UQFF term derived from thermodynamic infall ram pressure** rather than from electromagnetic, erosion, or tidal competition.

3. Compressed UQFF Form

The 13-term MUGE for NGC 1275 compresses to:

$$g_{\text{NGC1275}}(r, t) = g_{\text{MUGE10}}(r, t) + g_{\text{buoy}}^{\{(3)\}}(r, t)$$

where g_{MUGE10} contains the 10 original terms (base+BH+Ug+ Λ +EM+quantum+fluid+osc+DM+cooling), and:

$$g_{\text{buoy}}^{\{(3)\}} = \underbrace{0.5 \cdot \text{ug1}}_{\text{T1}} \cdot \underbrace{[-\beta_i \cdot \text{ug1} \cdot \omega_g \cdot \frac{M}{r} \cdot U_{\text{UA}} \cdot \cos(\pi t)]}_{\text{T2}} \cdot \underbrace{[-\beta_i \cdot \text{ug1} \cdot \omega_g \cdot \frac{M_{\text{vc}}}{r_{\text{vc}}} \cdot U_{\text{UA}} \cdot \cos(\pi t)]}_{\text{T3}}$$

The **AGN Feedback Equilibrium Tensor** (AFET) is then:

$$\mathcal{E}_{\text{AGN}} = \frac{\text{term}_{\text{cool}}}{\Sigma_{\text{buoy}}} = \frac{\rho_{\text{cool}} \cdot v_{\text{cool}}^2}{\rho_{\text{fluid}}} \cdot \text{ug1_base} \cdot [0.5 - \beta_i \cdot \omega_g \cdot \left(\frac{M}{r} + \frac{M_{\text{vc}}}{r_{\text{vc}}} \right) \cdot U_{\text{UA}} \cdot \cos(\pi t)]$$

At $\Sigma_{\text{AGN}} = 1$: equilibrium (self-regulated feedback) At $\Sigma_{\text{AGN}} > 1$: cooling-dominated $\rightarrow$ gas accumulation $\rightarrow$ AGN trigger At $\Sigma_{\text{AGN}} < 1$: buoyancy-dominated $\rightarrow$ quenched cooling $\rightarrow$ AGN quiescence

4. Observational Predictions

X-ray cavity regularity: The UQFF predicts quasi-periodic cavity pairs with period $\approx 2\pi/\omega_g = 2$ in natural units $\rightarrow$ corresponds to ~ 10 Myr intervals (consistent with Fabian et al. 2011 Perseus inventory of 12 cavity pairs).

Cooling flow suppression factor: At equilibrium, the net infall rate is suppressed by a factor $1/(1 + |\Sigma_{\text{buoy}}|/\text{term}_{\text{cool}})$ $\rightarrow$ predicts $\dot{M}_{\text{cool}}$ reduction from the classical value of $\sim 200 M_{\text{sun}} \text{ yr}^{-1}$ to the observed $\sim 30\text{--}50 M_{\text{sun}} \text{ yr}^{-1}$, a factor of 4–7 reduction. The UQFF buoyancy terms collectively provide this suppression.

Filament velocity distribution: The UQFF cosine modulation (Tier-2 and Tier-3) predicts filament infall velocities oscillate with a characteristic frequency $\omega_g = 7.3 \times 10^{-1} \text{ rad/s}$ $\rightarrow$ period ≈ 272 Myr. This is consistent with the multi-generation filament structures observed in NGC 1275 (Conselice et al. 2001, filament ages spanning $\sim 100\text{--}500$ Myr).

Virgo outer frame signature: The Tier-3 buoyancy uses the Virgo Cluster at ~ 77 Mpc as the outer gravitational frame. This predicts a **tidal contribution** to the ICM pressure profile at the cluster outskirts, potentially detectable as a departure from standard β -model fits at $r > 500$ kpc.

5. Significance

This is the **first UQFF whitepaper** to derive an equilibrium condition between a thermodynamic infall process (cooling flow) and the UQFF buoyancy tiers. It demonstrates:

The UQFF buoyancy framework is not merely additive — it creates a **dynamically self-regulating pair** with any infall/dissipative term that shares the same ug1_base kernel.

The AGN feedback cycle in BCGs is not fundamentally a thermodynamic cycle — it is a **gravitational field modulation cycle** governed by the UQFF buoyancy response to $G \cdot M/r^2$.

The Virgo Cluster outer frame (independent of Perseus at 77 Mpc) introduces a **super-cluster gravitational environment** into the local feedback physics — a prediction unique to UQFF multi-scale architecture.

UQFF computed: Eddington luminosity UQFF correction = $1 - [\text{SSq}] \exp(-\tau) = 1 - 5.7 \times 10^{-1} \exp(-2.9 \times 10^{-4}) = 4.3 \times 10^{-1}$; F_U at event horizon = $2.0 \times 10^{18} \text{ m/s}$.

References

- NGC1275.cpp (UQFF 2.0 upgrade, Session 72f, March 16, 2026) — $\text{term}_{\text{cool}} = \rho_{\text{cool}} * v_{\text{cool}}^2 / \rho_{\text{fluid}}$
- Fabian et al. (2011) — Perseus cluster: 12 X-ray cavity pairs, quasi-periodic AGN feedback
- McNamara & Nulsen (2007) — Heating of hot atmospheres with AGN jets
- Sanders & Fabian (2007) — Perseus cooling flow rate $\dot{M}_{\text{cool}} \sim 30\text{--}50 M_{\text{sun}} \text{ yr}^{-1}$

Taylor et al. (2006) — Perseus cluster ICM magnetic field $B \sim 5 \mu\text{G}$
Peterson & Fabian (2006) — X-ray spectroscopy of cooling clusters: cooling flow problem
Conselice et al. (2001) — NGC 1275 filamentary nebula structure and ages
CondensedPhysics3.py — NGC1275FBHFIlamentCalculator (PAPER_223, Session 56)
Star-Magic UQFF v4.26 — CP3/PAPER_198 3-tier buoyancy canonical framework

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